A σ-Essential Contextual State: Localizing an Open Problem on Concrete Orthomodular Lattices

Patrick S. Mahon*

Abstract

A two-valued state on a concrete orthomodular lattice (OML) of subsets of a set Ω assigns a definite truth value to each measurable event, coherently across overlapping measurement contexts. On a Boolean algebra every finitely coherent local pattern of such assignments extends to a global σ-additive two-valued state, and (away from a measurable cardinal) every global one is a point evaluation. We ask whether the non-distributive case can differ: does there exist a concrete σ-complete non-Boolean OML carrying a finitely coherent local pattern that extends to *no* global σ-additive two-valued state — a *σ-essential contextual state*? We show this question *localizes*: a witness exists iff (i) no point evaluation extends the pattern and (ii) no non-principal σ-additive two-valued state does either, and that clause (i) is freely arrangeable (after Navara–Pták), so the entire content sits in clause (ii). We then strip clause (ii) of all lattice scaffolding to a bare set-theoretic existence statement — a non-principal, σ-additive, coherent selection across a system of overlapping countable partitions — and locate it precisely against the surrounding literature: it is empty in the Polish-representable case (Derr–Williamson), it does *not* reduce to the Hilbert-space pure-state dichotomy of Blecher–Weaver (Kochen–Specker; Akemann–Weaver), and it does not reduce to a measurable cardinal in the way its Boolean shadow would. The result is a sharply posed, currently open problem at the interface of quantum logic and the set theory of measure-theoretic structures, together with a map of where its answer must live.

1 Introduction

A finite system of measurement contexts may admit a consistent assignment of truth values to events that no single underlying state realises — the algebraic signature of quantum contextuality. When the contexts are countable and one demands countable additivity, a sharper question arises: can a *finitely* coherent pattern fail to extend to any global σ-additive state, with the failure visible only at the σ-level? This paper isolates that phenomenon for concrete orthomodular lattices, reduces it to an irreducible set-theoretic core, and locates that core against the measure-theoretic and operator-algebraic literature. We work throughout with two-valued states, the non-distributive analogue of a $\{0,1\}$-valued measure.

A *concrete σ-complete orthomodular lattice* (concrete σ-OML) is a family $\mathcal{L} \subseteq \mathcal{P}(\Omega)$ with

$$\varnothing, \Omega \in \mathcal{L}; \qquad A \in \mathcal{L} \implies A^{\perp} := \Omega \setminus A \in \mathcal{L}; \qquad \{A_n\} \subseteq \mathcal{L} \text{ pairwise disjoint} \implies \bigcup_n A_n \in \mathcal{L},$$

ordered by inclusion. Equivalently it is a *σ-class* [17], a Gudder concrete quantum logic [10, 18]: the maximal Boolean subalgebras ("blocks") overlap but their union need not be Boolean. A *two-valued state* is $s\colon \mathcal{L} \to \{0,1\}$ with $s(\Omega) = 1$ that is additive over orthogonal joins lying in its domain: whenever $\{A_n\} \subseteq \mathcal{L}$ are pairwise disjoint with $\bigcup_n A_n \in \mathcal{L}$,

$$s(\textstyle\bigcup_n A_n) = \sum_n s(A_n)$$

*Research Computing Group, Simon Fraser University, Burnaby, BC, Canada. pmahon@sfu.ca

(in particular $s(A) + s(A^\perp) = 1$ on complement pairs); it is *σ-additive* when this holds for countable such families. The same additivity is required of states on a sub-orthoposet $B \subseteq \mathcal{L}$, over orthogonal joins that exist *in* B. The *point evaluations* $\delta_\omega(A) := \mathbb{1}[\omega \in A]$ are σ-additive two-valued states; call these *Dirac* and all others *non-Dirac*. For a finite $\perp$-closed sub-orthoposet $B \subseteq \mathcal{L}$ and a two-valued state s_0 on B, s_0 *extends* to s on $\mathcal{L}$ if $s \restriction B = s_0$.

Definition 1.1 (σ-essential contextual state)**.** A *σ-essential contextual state* on $\mathcal{L}$ is a two-valued state s_0 on a finite $\perp$-closed sub-orthoposet $B \subseteq \mathcal{L}$ that extends to *no* global σ-additive two-valued state on $\mathcal{L}$. A concrete σ-OML $\mathcal{L}$ *carries a σ-essential contextual state* (is a *witness*) if such an s_0 exists. We require $\mathcal{L}$ to be non-Boolean and *irreducible* (centre $\{\varnothing, \Omega\}$); see Remark 1.2.

The phenomenon is a failure of σ-level compactness. Wright's theorem [20] — every two-valued state on a *finite* OML is finitely realised — forbids any *finite* witness, so the question is whether a coherent finite pattern can fail to globalise once σ-additivity is imposed. On a Boolean algebra it cannot (§2); we isolate what must differ in the non-distributive case, reduce to the irreducible core, and locate it.

Results. §2: the Boolean baseline (Prop. 2.1, two obstructions). §3: the *Localization Theorem* (Thm. 3.2) — a witness exists iff (i) no Dirac and (ii) no non-Dirac σ-additive two-valued state extends s_0, with (i) freely arrangeable, localising all weight to (ii). §4: clause (ii) stripped to a bare set-theoretic selection problem (Q. 4.1). §5: the boundary map — empty above the Polish case, irreducible to Blecher–Weaver, irreducible to a measurable cardinal. §6–§7: the open problem and its forcing target.

Remark 1.2 (Irreducibility is not vacuous)**.** The carrier class is inhabited: the horizontal sum MO_κ (κ orthogonal atom-pairs with $0, 1$) is concrete, non-Boolean, and irreducible (centre $\{\varnothing, \Omega\}$) for $\kappa \geq 2$ [13], and σ-complete for $\kappa = \omega$. Concreteness does not force a non-trivial centre: by the Zierler–Schlessinger/Gudder criterion it is exactly order-determination by two-valued states [10, 11], unrelated to reducibility. (MO_κ itself is excluded as a witness by Remark 5.2.)

2 The Boolean baseline

We begin with the baseline the non-distributive case must escape: on a Boolean σ-algebra, Definition 1.1 is never satisfied. The proof isolates two distinct obstructions, both of which a witness must overcome at once.

Proposition 2.1 (No witness in the Boolean case)**.** *Let $\mathcal{L} \subseteq \mathcal{P}(\Omega)$ be a Boolean σ-algebra and $B \subseteq \mathcal{L}$ a finite $\perp$-closed sub-orthoposet with a two-valued state s_0. Then s_0 extends to a global σ-additive two-valued state on $\mathcal{L}$.*

Proof. Let $B_0 \subseteq \mathcal{L}$ be the finite Boolean subalgebra generated by B. Because $\mathcal{L}$ is Boolean and s_0 is additive over the orthogonal joins in B, s_0 extends to a (finitely additive) two-valued state on B_0: on a finite Boolean algebra such a state is determined by, and concentrates on, a single atom $a \in B_0$ — the meet of the s_0-true generators. As a is a nonzero element of the concrete algebra B_0, it is a nonempty subset of Ω, and

$$K(s_0) := \bigcap \{A \in B : s_0(A) = 1\} \supseteq a \neq \varnothing$$

(we call $K(s_0)$ the *kernel* of s_0). Any $\omega \in K(s_0)$ satisfies $\delta_\omega \restriction B = s_0$ by Lemma 3.1, and δ_ω is σ-additive. So s_0 globalises — and on a concrete Boolean σ-algebra it does so by a Dirac, the case $K(s_0) = \varnothing$ never arising. The extension to B_0 is exactly where distributivity is used; it is unavailable on a non-Boolean carrier, where the s_0-true generators may have empty meet. $\square$

Two distinct facts combine here, and a witness must defeat both *at once*:

(a) *(used in the proof)* when the events of B are *compatible*, they generate a finite Boolean algebra whose atom carrying s_0 is a nonempty common refinement — a common point. A witness breaks this: the events of B lie in incompatible blocks with empty meet, so no Dirac extends s_0.

(b) *(the complementary fact)* away from a measurable cardinal, every global σ-additive two-valued state on a Boolean σ-algebra is itself Dirac [19, 12]. So even where (a) is broken, on a Boolean carrier no *non*-Dirac state can repair the gap; a witness must ensure the gap stays irreparable on a non-distributive carrier where non-Dirac states are available.

The interplay is the crux: breaking (a) requires incompatibility (non-distributivity), yet that same non-distributivity is what makes (b) escapable, by admitting non-Dirac global states that might repair the gap. A witness must break (a) while keeping *no* such repair available.

3 The Localization Theorem

We first record the elementary characterisation of when a Dirac extends s_0.

Lemma 3.1. *Let $B \subseteq \mathcal{L}$ be a finite $\perp$-closed sub-orthoposet with two-valued state s_0. A point evaluation δ_ω satisfies $\delta_\omega \restriction B = s_0$ iff $\omega \in K(s_0) = \bigcap\{A \in B : s_0(A) = 1\}$. Consequently no Dirac extends s_0 iff $K(s_0) = \varnothing$.*

Proof. $\delta_\omega \restriction B = s_0$ means $\omega \in A \iff s_0(A) = 1$ for every $A \in B$. The forward constraints place $\omega \in \bigcap\{A : s_0(A) = 1\}$. For a false A (i.e. $s_0(A) = 0$), $\perp$-closure gives $A^\perp \in B$, and additivity gives $s_0(A^\perp) = 1 - s_0(A) = 1$; so $A^\perp$ is one of the true elements and the constraint $\omega \notin A$ is already $\omega \in A^\perp$, subsumed in $K(s_0)$. Thus the false constraints add nothing, and $\delta_\omega \restriction B = s_0 \iff \omega \in K(s_0)$. Conversely any $\omega \in K(s_0)$ works: true A give $\omega \in A$, false A give $\omega \in A^\perp$ hence $\omega \notin A$. □

Theorem 3.2 (Localization)**.** *Let $\mathcal{L}$ be a concrete σ-OML, $B \subseteq \mathcal{L}$ a finite $\perp$-closed sub-orthoposet, and s_0 a two-valued state on B. Then s_0 is a σ-essential contextual state iff*

(i) *$K(s_0) = \varnothing$ (no point evaluation extends s_0), and*

(ii) *no* non-Dirac *σ-additive two-valued state on $\mathcal{L}$ extends s_0.*

Moreover clause (i) *is freely arrangeable: there is a concrete σ-OML and an s_0 with $K(s_0) = \varnothing$.*

Proof. Every global σ-additive two-valued state on $\mathcal{L}$ is either a Dirac δ_ω or non-Dirac, and this dichotomy is exhaustive. Hence "no global σ-additive two-valued state extends s_0" is equivalent to the conjunction "no Dirac extends s_0" and "no non-Dirac extends s_0." By Lemma 3.1 the first conjunct is (i); the second is (ii). For arrangeability of (i), the Navara–Pták construction (Example 3.4) realises $K(s_0) = \varnothing$ explicitly. □

Remark 3.3 (Scope: a decomposition, not a reduction)**.** The equivalence is a decomposition, not a reduction to a separate named problem: the Dirac/non-Dirac split being exhaustive, it merely unfolds the definition of witness. Its content is that clause (i) is *freely arrangeable*, so all weight of Definition 1.1 sits in clause (ii) — the *σ-point-selection core*, the subject of the remainder.

The Navara–Pták construction both supplies clause (i) and shows (ii) is binding.

Example 3.4 (Navara–Pták [17]). Let $\Omega = \mathbb{Q}_0 \times \mathbb{Q}_0$ with $\mathbb{Q}_0 = \mathbb{Q} \cap (0,1)$, and let $f, g\colon \Omega \to \mathbb{R}$ be the coordinate projections. Let $\mathcal{L} = \mathcal{L}_{f,g}$ be the least σ-class containing the inverse images of Borel sets under f and g. Set $C_f = f^{-1}(\{1/3\})$, $C_g = g^{-1}(\{1/2\})$, $C_{f+g} = (f+g)^{-1}(\{1/2\})$. Define $m(A) = 1$ iff A contains one of C_f, C_g, C_{f+g}. Navara and Pták show m is a (two-valued, σ-additive) state on $\mathcal{L}$, and that since $C_f \cap C_g \cap C_{f+g} = \varnothing$ the state m is *not* concentrated at any point.

Remark 3.5 (The example is not itself a witness). Let B be the $\perp$-closure of $\{C_f, C_g, C_{f+g}\}$ and let s_0 be the pattern assigning each of the three the value 1. Since $C_f \cap C_g \cap C_{f+g} = \varnothing$, we have $K(s_0) = \varnothing$, so clause (i) holds — no point evaluation extends s_0. Yet the Navara–Pták state satisfies $m \restriction B = s_0$ and is non-Dirac and σ-additive, so clause (ii) *fails*, and s_0 is not σ-essential. The construction *builds its own rescuer*, which is why clause (i) can never suffice on its own. The reason the repair is available is that $\mathcal{L}_{f,g}$ is generated by two *compatible* (commuting) observables and is therefore Boolean enough to support a non-Dirac global state through the point-level gap. A witness demands the opposite: a carrier on which the point-level obstruction *cannot* be repaired by any non-Dirac state.

4 The bare statement

To see what is genuinely at stake in clause (ii), we strip away the lattice scaffolding and restate the core as a problem about selections. A two-valued state amounts to selecting one "true" cell within each block — a block being a maximal compatible context, equivalently a countable partition of Ω into elements of $\mathcal{L}$. Three conditions then encode the structure: overlapping blocks must agree on shared events (coherence); σ-additivity constrains the selection over countable joins; and non-Diracness means no single point realises it.

Question 4.1 (σ-point-selection, bare form). Given a set Ω and a family $\{\Pi_\alpha\}$ of countable partitions of Ω (the blocks of a concrete σ-OML, pairwise overlapping in their common refinements), does there exist a selection assigning to each Π_α one cell $c_\alpha \in \Pi_\alpha$ such that

(a) (coherence) the selections agree on events common to several blocks;

(b) (σ-additivity) the induced $\{0,1\}$-valuation commutes with countable disjoint unions;

(c) (non-principality) there is no $\omega_0 \in \Omega$ with $c_\alpha =$ the cell of Π_α containing ω_0 for all α;

(d) (prescription) the selection realises a prescribed finite pattern that no single point realises?

A witness exists iff Question 4.1 has a positive answer for some admissible carrier. In this bare form two features become legible that the lattice language obscures — its distance from a measurable cardinal, and its place within contextuality.

Remark 4.2 (The Boolean shadow is a measurable cardinal; the question is not). For a *compatible* block family (single Boolean base), a coherent non-principal σ-additive $\{0,1\}$-selection = a non-principal countably-complete ultrafilter = a measurable cardinal [12]. Question 4.1 imposes coherence (a) across *incompatible, mutually non-distributive* blocks — strictly stronger than an ultrafilter on any single block. So the *Boolean shadow* reduces to a measurable cardinal, but the non-distributive coherence is exactly what the reduction cannot cross (§5).

Remark 4.3 (Relation to sheaf-theoretic contextuality). Question 4.1 asserts that locally coherent $\{0,1\}$-valuations admit no *global section* — contextuality in the sense of Abramsky–Brandenburger [2], σ-additive form [1]. The framing is standard; what is open is the conjunction the literature does not treat together: countable additivity + a non-Hilbert (abstractly concrete) carrier + a $\{0,1\}$-valued selection.

5 The boundary map

We locate Question 4.1 by recording where its answer is known and where the obvious bridges fail.

The Polish boundary: empty above it. Derr and Williamson [6] prove a σ-extension criterion for coherent probabilities on pre-Dynkin systems: when the carrier is *Polish-representable* — Ω Polish, the generated σ-class equal to the Borel σ-algebra, with inner-regular blocks — finite coherence implies σ-extendability (their Theorem D.6, via Maharam [16]). Specialised to two-valued states, this says that on a Polish-representable carrier every finitely coherent s_0 globalises: *no σ-essential contextual state exists.* Hence a witness, if one exists, must live on a carrier that is *not* Polish-representable. This is the upper boundary of the problem.

The Hilbert boundary: an analogue that does not transfer. Blecher–Weaver [5]: on the projection lattice of $B(\ell^2(\kappa))$, a singular countably additive *pure* state exists iff κ is Ulam measurable — the nearest positive precedent that countable additivity on a non-distributive σ-complete OML *can* carry set-theoretic strength. It does not settle Question 4.1; three obstructions point the same way.

(1) *Wrong carrier.* For $\dim H \geq 3$ the projection lattice of $B(H)$ admits *no* two-valued state (Kochen–Specker [14]), so it is not concrete; the Blecher–Weaver dichotomy concerns *pure* states, not two-valued ones, which are absent there by Gleason [9].

(2) *Masa transfer fails.* Carrying a state-existence result to an abelian subalgebra means factoring through a masa; for the pure two-valued case this fails — Akemann–Weaver [3] give a pure state on $B(H)$ multiplicative on no masa, and Blecher–Weaver use Marcus–Spielman–Srivastava paving for exactly this reason.

(3) *The consumed object is structurally absent.* Blecher–Weaver extract their cardinal from a genuine ultrafilter on the diagonal masa $\ell^\infty(\kappa)$ (projections = subsets, intersection-closed). On a concrete non-Boolean σ-OML, $\mathcal{F}_s := \{A \in \mathcal{L} : s(A) = 1\}$ is an ultrafilter for *no* two-valued state s — not even a filter: meet-closure fails already for δ_ω, since in any concrete MO_2 the atoms a, b have $a \cap b \neq \varnothing$ (else orthogonal) yet $a \wedge b = 0$, so

$$\delta_\omega(a) = \delta_\omega(b) = 1, \qquad \delta_\omega(a \wedge b) = 0.$$

The countably-complete ultrafilter an Ulam-style argument would consume is absent from the carrier — the same non-distributivity (lattice meet $\neq$ set intersection), one level down.

Remark 5.1 (Strength of the claim)**.** Remark 4.2 and (1)–(3) are *cited facts assembled*, not a non-reducibility theorem in a fixed reduction calculus. The claim is only that the two obvious bridges — “measurable cardinal” and “transfer from Blecher–Weaver via a masa” — do not close Question 4.1, the obstruction in each being the same non-distributivity. This is what leaves the question open rather than a corollary.

Below: faithful representation forces distributivity. Routing Question 4.1 through a faithful σ-tribe representation (embedding $\mathcal{L}$ into a σ-algebra of sets, so two-valued states become point evaluations) is unavailable: a faithful tribe-of-sets representation forces distributivity, hence Booleanness (the “Floor” [18]). σ-Loomis–Sikorski holds under the Riesz Decomposition Property (σ-complete MV-algebras, RDP effect algebras [7]), but OMLs lack RDP (MO_2 witnesses), and the available image is only σ-epimorphic, not point-separating. The trivialising route is blocked, again non-distributively.

Remark 5.2 (Segregated carriers do not witness)**.** A carrier is *segregated* when its contextuality is carried by a central Boolean factor — e.g. MO_ω or the product $\prod_n MO_2$, whose σ-additive two-valued states all concentrate at points. On such carriers every finitely coherent s_0 is the restriction of a Dirac, so clause (ii) of Theorem 3.2 fails and there is no witness. A witness must be *non-segregated*: the incompatibility must be irreducible, not routed through a central partition. This is why Definition 1.1 requires irreducibility, and it sharpens the search for a carrier to the non-segregated, non-Boolean, non-Polish-representable regime.

6 The open problem

Assembling §3–§5, a σ-essential contextual state exists iff Question 4.1 has a positive answer on a carrier that is concrete, σ-complete, non-Boolean, irreducible, non-segregated, and non-Polish-representable. The boundary map of §5 settles the answer in no direction: none of the three natural bridges — the Blecher–Weaver dichotomy, a measurable cardinal, or a faithful representation — closes it.

Question 6.1 (Main)**.** Does there exist, in ZFC or under a large-cardinal hypothesis, a concrete σ-complete non-Boolean irreducible OML carrying a σ-essential contextual state? Equivalently, is the existence of a non-principal coherent σ-additive $\{0,1\}$-selection across overlapping countable partitions (Question 4.1, non-segregated case) provable, refutable, or independent?

What a resolution would require:

Positive (ZFC).
A concrete construction across the non-Polish, non-segregated regime — necessarily globally non-compositional, since (Remarks 4.2, 5.1) no admissible carrier assembles from countably many compatible pieces.

Negative.
Extends Derr–Williamson's Polish-representable emptiness past its topological hypotheses to all concrete σ-OMLs — against the Blecher–Weaver evidence that the analogous Hilbert cell is inhabited under a large cardinal.

Independent.
A model with the selection and one without. Known precedents in the set theory of measure-theoretic structures (von Neumann–Maharam [4], real-valued measurable cardinals [8]) are all real-valued and strictly positive; the two-valued, non-distributive case is the open edge.

7 A forcing target

The shape of the boundary map suggests where a resolution of Question 6.1 should be sought: not in ZFC and not in a single named combinatorial principle, but in an independence proof. We record the target precisely, as a hand-off for set-theoretic methods.

We build the target sentence in layers, so that its quantifier structure — the feature that decides which set-theoretic methods apply — is visible. Write $\mathrm{St}(B)$ for the two-valued states on a sub-orthoposet B and $\mathrm{St}_\sigma(\mathcal{L})$ for the global σ-additive two-valued states on $\mathcal{L}$.

The atom: a lift. That a global state μ extends a local pattern s is the single equation

$$\mu\restriction B = s, \qquad \mu \in \mathrm{St}_\sigma(\mathcal{L}),\ s \in \mathrm{St}(B).$$

Layer one: s admits a lift. Fixing B and s, the assertion that some global σ-additive state realises s is an existential:

$$\mathrm{lift}(B,s) \;:\equiv\; \exists\, \mu \in \mathrm{St}_\sigma(\mathcal{L})\ \ \mu\restriction B = s.$$

Layer two: every pattern lifts. Quantifying over all finite $\perp$-closed B (write $B \in \mathrm{Fin}_\perp(\mathcal{L})$) and all patterns on them gives the *σ-additive lifting* property of the carrier:

$$\Phi(\mathcal{L}) \;:\equiv\; \forall B \in \mathrm{Fin}_\perp(\mathcal{L}) \;\; \forall s \in \mathrm{St}(B) \;\; \exists \mu \in \mathrm{St}_\sigma(\mathcal{L}) \;\; \mu \restriction B = s.$$

Layer three: a binding pattern. Negating exhibits a local pattern no global state can rescue — exactly a σ-essential contextual state (Def. 1.1, Thm. 3.2):

$$\neg\Phi(\mathcal{L}) \;\equiv\; \exists B \;\; \exists s \in \mathrm{St}(B) \;\; \forall \mu \in \mathrm{St}_\sigma(\mathcal{L}) \;\; \mu \restriction B \neq s.$$

Layer four: a witness exists. Quantifying over admissible carriers (concrete, σ-complete, non-Boolean, irreducible; write $\mathcal{L} \in \mathrm{Adm}$) yields the target sentence:

$$\Psi \;:\equiv\; \exists \mathcal{L} \in \mathrm{Adm} \;\; \neg\Phi(\mathcal{L}) \;\equiv\; \exists \mathcal{L} \in \mathrm{Adm} \; \exists B \; \exists s \;\; \forall \mu \in \mathrm{St}_\sigma(\mathcal{L}) \;\; \mu \restriction B \neq s.$$

Reading off the shape: Ψ is $\exists\exists\exists\forall$, and its leading quantifier ranges over carriers $\mathcal{L}$ — a second-order object, a family of subsets of Ω, not a natural number. By Proposition 2.1, Φ holds for every Boolean $\mathcal{L}$, so the witness $\mathcal{L}$ in Ψ is necessarily non-distributive. It is this leading second-order existential over an unbounded carrier, not the inner arithmetic, that places Ψ beyond a single combinatorial principle — as the two features below make precise.

Two features make Ψ a forcing target, not a construction target:

(1) *No compositional carrier.* Any countable, refinable approximation collapses to the Boolean case (Prop. 2.1, Rem. 5.2), so a witness must be uncountably generated and globally non-compositional.

(2) *No fixed carrier needed.* An independence proof need only produce models where Ψ holds and fails, the carrier supplied by the model — as for the von Neumann measure-algebra problem, settled by models, not a fixed algebra [4, 16].

Remark 7.1 (Candidate technology and conjectured strength)**.** The setting is the set theory of measure-theoretic and operator-algebraic structures; the nearest cardinal-explicit precedent, Blecher–Weaver [5], does not transfer (§5), so the strength of Ψ is *conjectural, not inherited.* Two caveats fix the technique:

- *Cardinal type.* If any, the relevant cardinal is Ulam-measurable, not real-valued measurable: Ψ is two-valued, where the $[0, 1]$-valued machinery does not apply.
- *Forcing frames, does not construct.* The bare object has, on a Boolean base, the strength of a measurable cardinal; set-forcing is consistency-strength preserving (does not create measurables, Lévy–Solovay [15]). So forcing can frame the *independence* but cannot force a witness into existence — the positive direction is a *direct* large-cardinal question, as Blecher–Weaver's threshold (an equivalence, not a construction) indicates.

All known precedents [4, 8] are real-valued and strictly positive; the two-valued non-distributive case is, as far as we are aware, untreated. We state the strength as a question, not a theorem:

Is Ψ independent of ZFC, and if so, of what consistency strength?

Acknowledgements. The author thanks the anonymous referees (forthcoming) and acknowledges the use of large language model assistance (Anthropic Claude, OpenAI ChatGPT) for literature search, verification of cited statements against primary sources, and exposition; all mathematical claims were independently verified by the author.

References

[1] Samson Abramsky and Rui Soares Barbosa. The sheaf-theoretic structure of definite causality. *Communications in Mathematical Physics*, 391, 2022. σ-additive sheaf-theoretic contextuality; see also arXiv:1905.08267.

[2] Samson Abramsky and Adam Brandenburger. The sheaf-theoretic structure of non-locality and contextuality. *New Journal of Physics*, 13:113036, 2011.

[3] Charles A. Akemann and Nik Weaver. $\mathcal{B}(H)$ has a pure state that is not multiplicative on any masa. *Proceedings of the National Academy of Sciences*, 105(14):5313–5314, 2008.

[4] Bohuslav Balcar, Thomas Jech, and Tomáš Pazák. Complete ccc Boolean algebras, the order sequential topology, and a problem of von Neumann. *Bulletin of the London Mathematical Society*, 37(6):885–898, 2005.

[5] David P. Blecher and Nik Weaver. Quantum measurable cardinals. *Journal of Functional Analysis*, 273(5):1870–1890, 2017.

[6] Rabanus Derr and Robert C. Williamson. Systems of precision: coherent probabilities on pre-Dynkin-systems and coherent previsions on linear subspaces. *arXiv preprint arXiv:2302.03522*, 2023.

[7] Anatolij Dvurečenskij. Loomis–sikorski theorem for σ-complete MV-algebras and ℓ-groups. *Journal of the Australian Mathematical Society*, 68(2):261–277, 2000.

[8] David H. Fremlin. *Measure Theory, Volume 5: Set-theoretic Measure Theory.* Torres Fremlin, 2008.

[9] Andrew M. Gleason. Measures on the closed subspaces of a Hilbert space. *Journal of Mathematics and Mechanics*, 6(6):885–893, 1957.

[10] Stanley P. Gudder. Quantum probability spaces. *Proceedings of the American Mathematical Society*, 21:296–302, 1969.

[11] John Harding. Remarks on concrete orthomodular lattices. *International Journal of Theoretical Physics*, 43(10):2149–2168, 2004.

[12] Thomas Jech. *Set Theory.* Springer, 3rd millennium edition, 2003.

[13] Gudrun Kalmbach. *Orthomodular Lattices.* Academic Press, 1983.

[14] Simon Kochen and Ernst P. Specker. The problem of hidden variables in quantum mechanics. *Journal of Mathematics and Mechanics*, 17(1):59–87, 1967.

[15] Azriel Lévy and Robert M. Solovay. Measurable cardinals and the continuum hypothesis. *Israel Journal of Mathematics*, 5:234–248, 1967.

[16] Dorothy Maharam. Consistent extensions of linear functionals and of probability measures. In *Proceedings of the Sixth Berkeley Symposium on Mathematical Statistics and Probability, Vol. II: Probability Theory*, pages 127–147. University of California Press, Berkeley, 1972.

[17] Mirko Navara and Pavel Pták. Two-valued measures on σ-classes. *Časopis pro pěstování matematiky*, 108(3):225–229, 1983.

[18] Pavel Pták and Sylvia Pulmannová. *Orthomodular Structures as Quantum Logics*, volume 44 of *Fundamental Theories of Physics.* Kluwer, Dordrecht, 1991.

[19] Roman Sikorski. *Boolean Algebras.* Springer, 2nd edition, 1964.

[20] Ron Wright. The state of the pentagon: a nonclassical example. In A. R. Marlow, editor, *Mathematical Foundations of Quantum Theory*, pages 255–274. Academic Press, New York, 1978.